

Digital Electronics

Circuit Example: A Four-Bit Adder

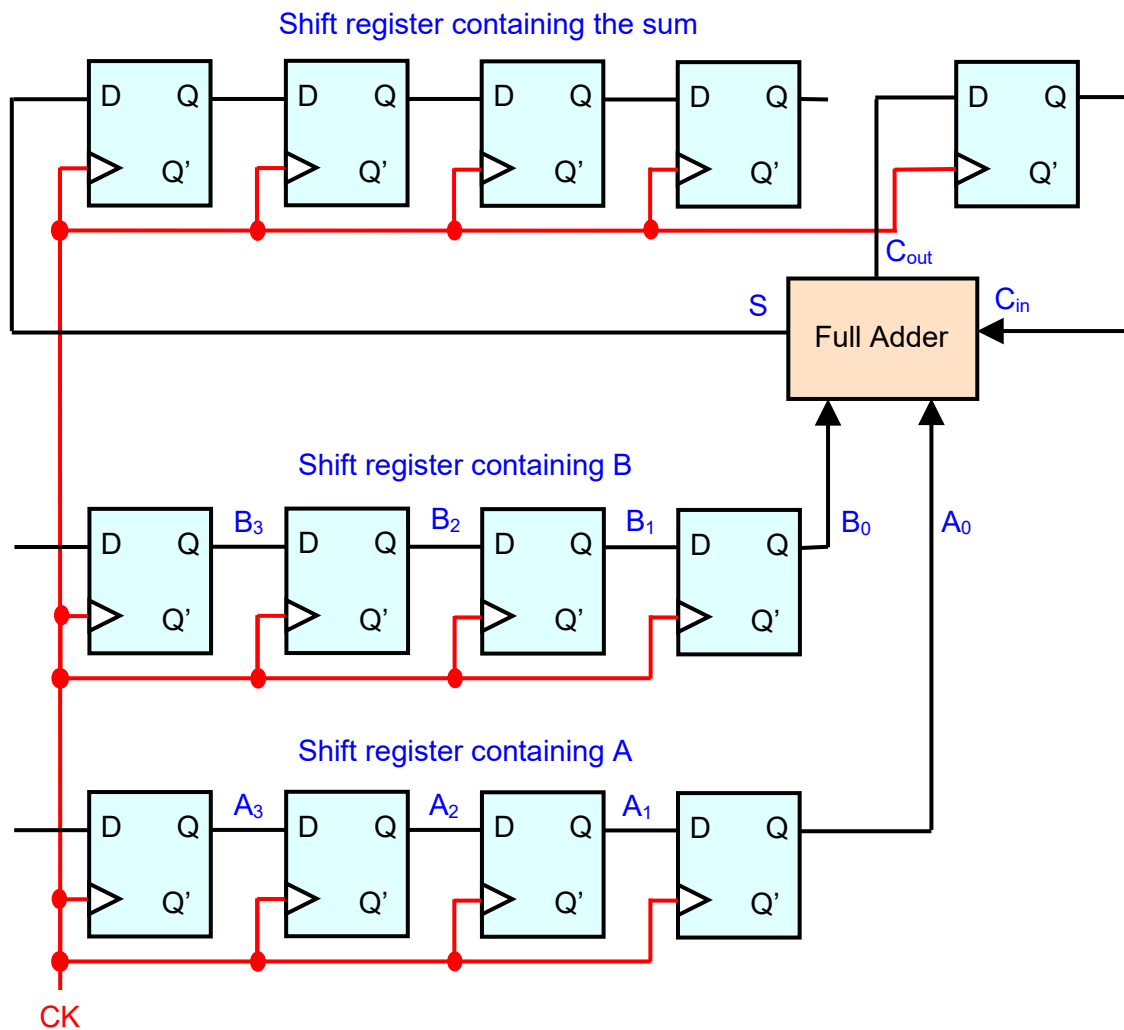
Using A Single Full Adder

S Le Goff

June 2026

6. Four-bit adder using a single full-adder

Design of a four-bit adder using a single full adder.



Several shift registers must be available to store the two four-bit numbers $A = (A_3 A_2 A_1 A_0)_2$ and $B = (B_3 B_2 B_1 B_0)_2$ as well as their sum $S = A + B$ and the carry generated at each stage.